

# MATH 3551 A — Abstract Algebra I

## Practice Midterm 1

Taylor Dupuy, University of Vermont • Fall 2026

For practice — Midterm 1 is Monday, October 12, 2026 • Time limit: 50 minutes • Total: 100 points

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NAME (please print): \_\_\_\_\_

**Directions.** This exam is **closed book**: no textbook, no notes, no calculators, no phones, no other devices, and no assistance of any kind. Write your answers on the exam itself; use the backs of pages if you need more room, and say clearly when you have done so. *Show your work and name the theorems you use* — “by Lagrange’s Theorem,  $|H|$  divides 24” earns nearly all of the credit, an unjustified correct number earns little. Point values are marked on each problem. Good luck.

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**About this practice exam.** It was drafted with the help of Claude Opus 5, running in a Cowork harness, working from previous exams in this course. **It has not yet been timed or proofread.** Like the real exam, it is built to take well under the time you are given: about thirty minutes of a fifty-minute midterm period, and about a hundred minutes of the final’s two hours forty-five. The spare time is deliberate. *Please hunt for mistakes.* If you find a wrong answer, an ambiguous or unanswerable problem, a problem that leans on something we have not covered, or one that takes far longer than its point value suggests, tell me — in class, in office hours, or by email. I would much rather hear it from you now than print it on the real exam.

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**Problem 1.** (30 points) **True or false; prove or disprove.**

Circle **T** or **F**, and then give a one-line reason: a theorem, a short computation, or an explicit counterexample. Five points each: two for the circle, three for the reason.

- (a) **T**   **F**     There is an element of order 18 in the symmetric group  $S_9$ .
- (b) **T**   **F**     The map  $\varphi: \mathbb{Q}^\times \rightarrow \mathbb{Q}^\times$  defined by  $\varphi(x) = |x|$  is a group homomorphism.
- (c) **T**   **F**     The center of the dihedral group  $D_{16}$  of order 16 is  $\{1, r^4\}$ .
- (d) **T**   **F**     The group  $(\mathbb{Z}/15\mathbb{Z})^\times$  is cyclic.
- (e) **T**   **F**     The number of 3-cycles in  $S_6$  is 40.

(f) **T**   **F**   The groups  $\mathbb{Z}/4\mathbb{Z}$  and  $(\mathbb{Z}/8\mathbb{Z})^\times$  are isomorphic.

**Problem 2.** (25 points) **Inside  $\mathbb{Z}/72\mathbb{Z}$ .**

Work in the additive group  $\mathbb{Z}/72\mathbb{Z}$ .

- (a) (4 points) Find the order of  $\overline{30}$ .
- (b) (5 points) List the elements of the subgroup  $\langle \overline{30} \rangle$ .
- (c) (4 points) How many generators does  $\mathbb{Z}/72\mathbb{Z}$  have?
- (d) (7 points) Determine every  $k$  with  $0 \leq k < 72$  for which  $\langle \overline{k} \rangle = \langle \overline{18} \rangle$ .

**Problem 3.** (25 points) **Two permutations.**

In  $S_9$  let

$$\sigma = (1\ 4\ 7)(2\ 5)(3\ 6\ 8\ 9), \quad \tau = (1\ 2\ 3\ 4\ 5).$$

(Permutations act on the left, so  $\sigma\tau$  means: first apply  $\tau$ , then apply  $\sigma$ .)

- (a) (4 points) What is  $|\sigma|$ ?
- (b) (6 points) Find the cycle decomposition of  $\sigma\tau$ , and its order.
- (c) (3 points) For which  $k$  with  $0 \leq k < 12$  does  $\sigma^k$  fix the point 3?
- (d) (2 points) Give the cycle decomposition of  $\sigma^{-1}$ .

**Problem 4.** (20 points) **Centralizer and normalizer in the quaternion group.**

Let  $G = Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ , with the usual relations  $i^2 = j^2 = k^2 = -1$ ,  $ij = k$ ,  $ji = -k$ , and let  $A = \langle i \rangle = \{1, -1, i, -i\}$ .

- (a) (6 points) Compute  $C_G(A)$ . Justify your answer.
- (b) (6 points) Compute  $N_G(A)$ . Justify your answer.
- (c) (3 points) List the left cosets of  $A$  in  $G$  and check Lagrange's Theorem for this pair.